

Current Prediction in Analog Integrated Circuits Using Graph Attention Networks - Update

Matthew Jones
Oregon State University
Corvallis, Oregon
jonesm25@oregonstate.edu

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1 Introduction

This is an update based on the Future Work section of the previous report. Please refer to the it for details.

2 Per-Process Model Performance

The GAN was trained independently on 10^5 Monte Carlo simulations of the 2-inverter circuit for each of the nine PTM process models. All runs used the same architecture, learning-rate schedule, and 100-epoch budget. The W&B chart is shown in Figure 1. Final metrics are summarized in Table 1. In every run the training loss and average test loss are within $\sim 1\%$ of each other, indicating the model is not overfitting. The cross-process variation is driven by representation and target-distribution effects rather than generalization gap.

Process	Train Loss	Test Loss	MRE
22nm_LP	0.157	0.153	0.75
22nm_HP	0.221	0.220	1.84
32nm_LP	0.118	0.120	1.27
32nm_HP	0.187	0.189	3.22
45nm_LP	0.116	0.116	2.64
45nm_HP	0.202	0.201	9.06
65nm_bulk	0.178	0.180	2.93
90nm_bulk	0.236	0.236	9.72
130nm_bulk	0.306	0.306	36.72

Table 1: Per-process performance for each process (MRE = mean relative error)

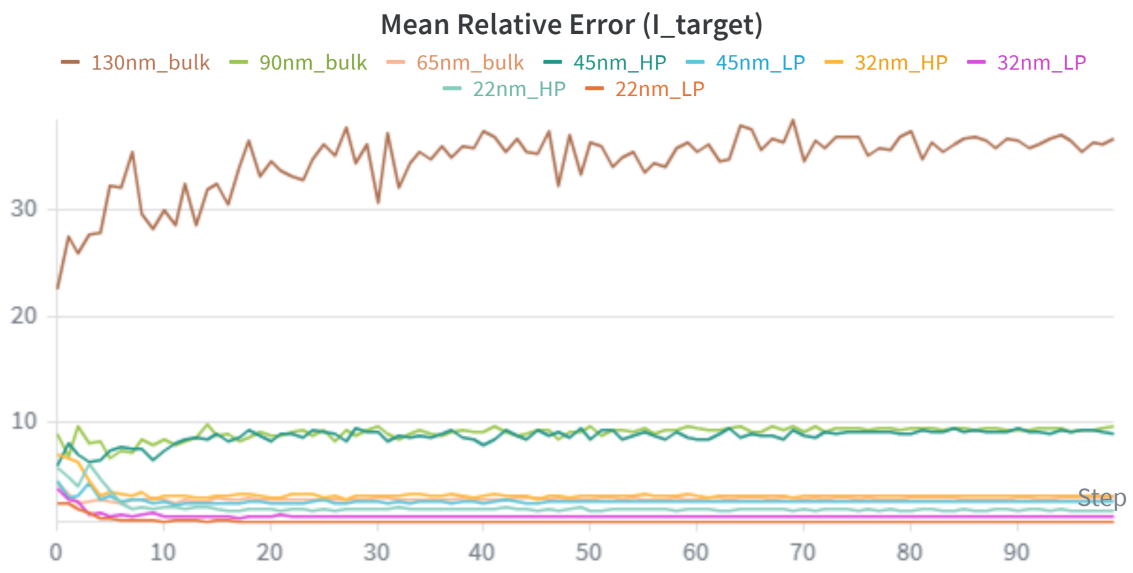


Figure 1: Mean Relative Error over 100 epochs